Examiner only

11. Carboxylic acids are classed as weak acids.

(a) State what is meant by the term weak acid. [1]

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(b) Substituted propanoic acids, $CH_3CHXCOOH$, have different acid strengths. The table below gives some information about three substituted propanoic acids.

Substituted propanoic acid, $CH_3CHXCOOH$	
X = H	The pH of a 0.50 mol dm^{-3} aqueous solution is 2.59
X = Cl	$K_a = 1.48 \times 10^{-3}$ mol dm^{-3}
X = OH	pK_a = 3.86

Place these acids in order of increasing acid strength. **You must show your working**. [3]

Weakest .... Strongest

Examiner only

(c) A buffer is produced by mixing $100\,cm^3$ of ethanoic acid ($K_a = 1.8 \times 10^{-5}\,mol\,dm^{-3}$) of concentration $0.80\,mol\,dm^{-3}$ with $50\,cm^3$ of aqueous sodium hydroxide of concentration $0.80\,mol\,dm^{-3}$ at a temperature of 298 K.

(i) Calculate the initial pH of the aqueous sodium hydroxide. [2]

pH = ...

(ii) Calculate the pH of the buffer solution. [2]

pH = ...

(iii) Explain how buffers such as this work. [3]

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